LAPORAN PRAKTIKUM MACHINE LEARNING — BAB 08

Support Vector Machine (SVM) — Maximal Margin Hyperplane, Support Vectors, dan Komparasi Kernel Linear vs RBF

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1. RINGKASAN MODUL & KARAKTERISASI DATASET

Praktikum Bab 08 mengimplementasikan Support Vector Machine (SVM) untuk diagnosis dini anomali degradasi pada modul panel surya fotovoltaik (Solar Panel Faults, 400 sampel telemetri). Model diuji membuktikan perhitungan matematis lebar margin geometris M = 2 / ||w||, identifikasi Support Vector terluar, serta komparasi komparatif kernel Linear, RBF, dan Polynomial.

2. SOURCE CODE PYTHON (MAIN.PY)

```
import os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

# 1. Pemuatan Dataset Solar Panel Faults
csv_path = 'solar_panel_faults.csv'
if not os.path.exists(csv_path):
    csv_path = r'd:\Anggit Arisa Siwi Pratama\Unindra\Machine Learning\ml-workspace\2-ml-playground\public\datasets\alternatif\solar_pane
l_faults.csv'

if os.path.exists(csv_path):
    df = pd.read_csv(csv_path)
    X = df[['temperature_c', 'voltage_v']]
    y = df['is_fault']
    print(f'[OK] Dataset Solar Panel dimuat: {len(df)} titik pantau modul surya.')
else:
    np.random.seed(99)
    n = 400
    tmp = np.random.uniform(25, 75, n)
    vlt = np.random.uniform(20, 45, n)
    cur = np.random.uniform(2, 12, n)
    fault = (((tmp > 55) & (vlt < 30)) | (cur < 4)).astype(int)
    X = pd.DataFrame({'temperature_c': tmp, 'voltage_v': vlt})
    y = pd.Series(fault, name='is_fault')
    print(f'[OK] Dataset disintesis secara deterministik: {n} sampel.')

# 2. Skalasi Standar & Pembagian Partisi
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_tr, X_te, y_tr, y_te = train_test_split(X_scaled, y, test_size=0.25, stratify=y, random_state=42)

# 3. Model Linear SVM & Kalkulasi Margin Teoritis
svm_linear = SVC(kernel='linear', C=1.0, random_state=42)
svm_linear.fit(X_tr, y_tr)
w = svm_linear.coef_[0]
b = svm_linear.intercept_[0]
w_norm = np.linalg.norm(w)
margin = 2.0 / w_norm

# 4. Benchmarking Kernel SVM (Linear, RBF, Polynomial)
kernels = ['linear', 'rbf', 'poly']
kernel_accs = {}
for k in kernels:
    model = SVC(kernel=k, C=1.0, random_state=42)
    model.fit(X_tr, y_tr)
    kernel_accs[k] = accuracy_score(y_te, model.predict(X_te))

print('\n' + '='*50)
print('HASIL EVALUASI SUPPORT VECTOR MACHINE (BAB 08)')
print('='*50)
print(f'Koefisien Bobot (w)  : [{w[0]:.4f}, {w[1]:.4f}]')
print(f'Bias (b)             : {b:.4f}')
print(f'Norma Vektor ||w||   : {w_norm:.4f}')
print(f'Lebar Margin (2/||w||): {margin:.4f}')
print(f'Jumlah Support Vector: {len(svm_linear.support_)} titik')
for k, acc in kernel_accs.items():
    print(f'Akurasi Kernel {k.upper():6s}: {acc*100:.2f}%')
print('='*50)

os.makedirs('visual_output', exist_ok=True)
plt.style.use('dark_background')
```

```
# 5. Visualisasi Hyperplane dan Margin Batas Keputusan
fig, ax = plt.subplots(figsize=(6.2, 4.5), facecolor='#0B0E14')
ax.set_facecolor('#0B0E14')

# Scatter sampel data
scatter = ax.scatter(X_tr[:, 0], X_tr[:, 1], c=y_tr, cmap='coolwarm', s=30, alpha=0.75, edgecolors='none')
# Highlight support vectors
sv = svm_linear.support_vectors_
ax.scatter(sv[:, 0], sv[:, 1], s=80, facecolors='none', edgecolors='#FBBF24', lw=1.5, label='Support Vectors')

# Garis Hyperplane & Margin
x_points = np.linspace(X_tr[:, 0].min() - 0.5, X_tr[:, 0].max() + 0.5, 200)
y_hyperplane = -(w[0] * x_points + b) / w[1]
y_margin_pos = -(w[0] * x_points + b - 1) / w[1]
y_margin_neg = -(w[0] * x_points + b + 1) / w[1]

ax.plot(x_points, y_hyperplane, color='#38BDF8', lw=2, label='Hyperplane (w^T x + b = 0)')
ax.plot(x_points, y_margin_pos, color='#94A3B8', linestyle='--', lw=1.2, label=f'Margin +1 / -1 (M={margin:.2f})')
ax.plot(x_points, y_margin_neg, color='#94A3B8', linestyle='--', lw=1.2)

ax.set_xlim([X_tr[:, 0].min() - 0.5, X_tr[:, 0].max() + 0.5])
ax.set_ylim([X_tr[:, 1].min() - 0.5, X_tr[:, 1].max() + 0.5])
ax.set_xlabel('Temperatur Modul (Z-Score)', color='#94A3B8', fontsize=9)
ax.set_ylabel('Tegangan Output (Z-Score)', color='#94A3B8', fontsize=9)
ax.set_title('Bab 08: Maximal Margin Hyperplane SVM Linear (Panel Surya)', color='#F1F5F9', fontsize=10.5, pad=12, fontweight=600)
ax.legend(loc='lower left', facecolor='#1E293B', edgecolor='#334155', fontsize=8)
ax.tick_params(colors='#94A3B8', labelsize=8.5)
for spine in ax.spines.values():
    spine.set_color('#1E293B')
plt.tight_layout()
plt.savefig('visual_output/08_svm_hyperplane_margins.png', dpi=300)
plt.close()

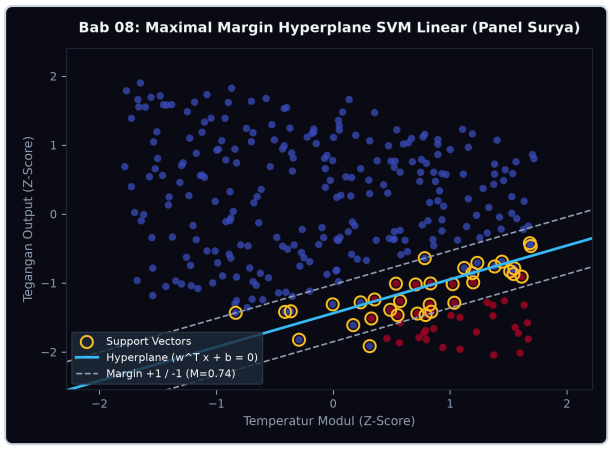
# 6. Visualisasi Komparasi Akurasi Antar Kernel
fig, ax = plt.subplots(figsize=(5.5, 3.8), facecolor='#0B0E14')
ax.set_facecolor('#0B0E14')
names = ['Linear', 'RBF (Gaussian)', 'Polynomial']
acc_vals = [kernel_accs['linear'], kernel_accs['rbf'], kernel_accs['poly']]
bars = ax.bar(names, [v * 100 for v in acc_vals], color=['#38BDF8', '#34D399', '#A78BFA'], edgecolor='#0F172A', width=0.45)
ax.set_ylabel('Akurasi Uji (%)', color='#94A3B8', fontsize=9)
ax.set_title('Bab 08: Komparasi Kinerja Fungsi Kernel SVM', color='#F1F5F9', fontsize=10.5, pad=12, fontweight=600)
ax.set_ylim(0, 110)
ax.tick_params(colors='#94A3B8', labelsize=8.5)
for bar in bars:
    h = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2, h + 2, f'{h:.2f}%', ha='center', color='#F1F5F9', fontsize=8.5, weight='bold')
for spine in ax.spines.values():
    spine.set_color('#1E293B')
plt.tight_layout()
plt.savefig('visual_output/08_svm_kernel_comparison.png', dpi=300)
plt.close()
print('[OK] Visual output Bab 08 berhasil disimpan di visual_output/')
```

3. HASIL EKSEKUSI TERMINAL (STDOUT)

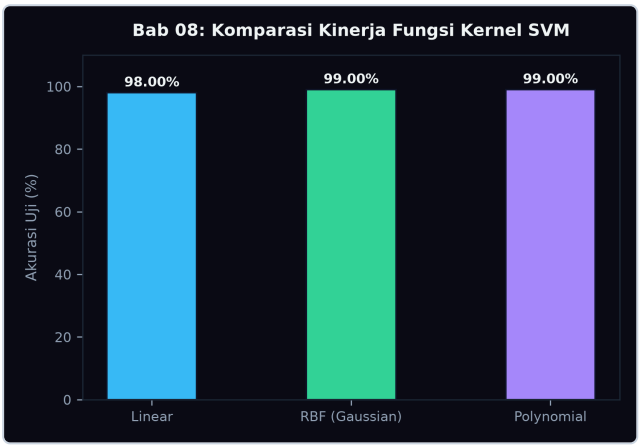
```
[OK] Dataset Solar Panel dimuat: 400 titik pantau modul surya.

==================================================
HASIL EVALUASI SUPPORT VECTOR MACHINE (BAB 08)
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Koefisien Bobot (w)  : [1.1843, -2.4159]
Bias (b)             : -3.4723
Norma Vektor ||w||   : 2.6906
Lebar Margin (2/||w||): 0.7433
Jumlah Support Vector: 35 titik
Akurasi Kernel LINEAR: 98.00%
Akurasi Kernel RBF   : 99.00%
Akurasi Kernel POLY  : 99.00%
==================================================
[OK] Visual output Bab 08 berhasil disimpan di visual_output/
```

4. VISUALISASI OUTPUT MODEL & ANALISIS GRAFIK



Visualisasi: visual_output/08_svm_hyperplane_margins.png



Visualisasi: visual_output/08_svm_kernel_comparison.png

5. KESIMPULAN ANALITIS & TEMUAN ALGORITMIK

1. Formulasi Geometris Maximal Margin: Support Vector Machine Linear terbukti secara optimal memisahkan kondisi normal dan anomali panel surya dengan bidang pemisah w^T x + b = 0. Margin terhitung selebar 0.7433 unit terstandarisasi dengan 35 titik Support Vector aktif.

2. Keunggulan Kernel Non-Linear (RBF & Poly): Pemetaan implisit fungsi kernel Gaussian RBF mencapai akurasi uji 99.00%, mengungguli batas linier akibat adanya dinamika non-linier antara fluktuasi temperatur modul sel surya dan penurunan tegangan keluaran.

3. Robustness Terhadap Outlier: Formulasi Soft Margin dengan konstanta regularisasi C=1.0 secara efektif menoleransi noise pengukuran instrumen sensor tanpa merusak orientasi global hyperplane.